

WEEKLY PROGRESS REPORT

Week 1: DNS Tunneling & Detection Basics

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Project: CyberSec Internship - Cygnus-Security
Related Issue: #7

1. Contribution Scope

This week's work contributes to the **Network Security & Monitoring** phase of the internship project. The primary focus is identifying covert communication channels and developing detection logic for the Intrusion Detection System (IDS) / Deep Packet Inspection (DPI) framework.

2. Objective of This Week

- Understand the core concept and mechanics of DNS tunneling.
- Identify common attack patterns and adversary techniques (Exfiltration, C&C).
- Learn basic to advanced detection approaches using network traffic analysis.
- Develop a proof-of-concept (PoC) tool for lexical anomaly detection.

3. Detailed Work

3.1 Description

DNS Tunneling is a stealthy evasion technique that weaponizes the DNS protocol (Port 53) to encapsulate malicious data within legitimate-looking queries. It bypasses Layer 3/4 firewalls by exploiting the inherent trust in DNS recursive resolvers.

3.2 Design / Method

The detection strategy relies on **Lexical Analysis** using **Shannon Entropy** to measure the randomness of subdomains. This is combined with **Protocol Feature Analysis** (inspecting TXT/NULL records) and **Traffic Volume Analysis** (detecting beaconing behaviors).

3.3 Implementation

Developed a Python-based detection engine (`dns_entropy_analyzer.py`) that isolates subdomains using the `tldextract` library and applies a Dual-Threshold logic (Length + Entropy) to trigger alerts.

4. Results / Output

- **Research Document:** Comprehensive analysis of DNS tunneling mechanics and tools (DNScat2, Iodine).
- **Detection Tool:** A functional Python script that flags high-entropy DNS queries.
- **Detailed Report:** Attached as *week1_dns-tunneling-research.pdf*.

5. Analysis

The heart of the detection logic is the **Shannon Entropy** formula, which measures the "information density" or randomness of a string:

$$H(X) = - \sum_{i=1}^k P(x_i) \log_2 P(x_i) \quad (1)$$

Where $P(x_i)$ is the probability of character x_i appearing in the subdomain. Encrypted or encoded malware payloads (Base64) typically exhibit significantly higher entropy scores compared to natural language domains.

6. Issues & Fixes

- **Issue:** High False Positive (FP) rate when analyzing the full domain string.
- **Fix:** Isolated the subdomain "payload container" by stripping the root domain and TLD.
- **Issue:** Legitimate long URLs (CDN, tracking links) triggering alerts.
- **Fix:** Implemented a **Dual-Engine Threshold:** Subdomain Length ≥ 15 AND Entropy ≥ 3.5 .

7. Reusable Content ★

The Domain Name System (DNS) remains a critical blind spot in perimeter defense. Because DNS is essential for network operations, Port 53 is typically left unmonitored by traditional firewalls. Adversaries exploit this by encoding non-DNS data into the subdomain fields of queries.

To counteract this, a robust detection architecture must employ Deep Packet Inspection (DPI) to extract Layer 7 metadata. Lexical analysis, specifically Shannon Entropy, provides a mathematical baseline for identifying encoded payloads. When a subdomain exhibits high entropy alongside unusual length, it indicates a high probability of data

exfiltration or C&C activity. Temporal analysis of inter-arrival times further identifies automated beaconing characteristic of malware scripts.

8. Next Step

- Integrate with Suricata's `eve.json` logs for real-time traffic analysis.
- Implement stateful tracking to monitor query frequency per IP address.